



UNIVERSITI TUNKU ABDUL RAHMAN

Tutorial 4

1. Describe three commonly used methods for performing economic cost-benefit analysis?
2. Assuming monetary benefits of an information system at RM 85,000 per year, one time costs of RM 75,000, recurring costs of RM 35,000 per year, a discount rate of 12 percent, and a five-year time horizon, calculate the net present value of these costs and benefits of an information system. Also, calculate the overall return on investment of the project and then present a break-even analysis. At what point does break even occur?

Problems							
	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Totals
Net Economic Benefit							
Discount Rate (%)							
PV of Benefits							
NPV of all Benefits							
One-time Costs							
Recurring Costs							
Discount Rate (%)							
PV of Recurring Costs							
NPV of all Costs							
Overall NPV							
Overall ROI							
Break-even Analysis							
Yearly NPV Cash Flow							
Overall NPV Cash Flow							
Project break even occurs between () and () years							
Use first year of positive cash flow to calculate break-even fraction							
Actual break even occurs at							

3. Assume monetary benefits of an information system of RM 50,000 the first year and increasing benefits of RM 5,000 a year for the next four years (Year 1 = 50,000; Year 2 = 55,000; Year 3 = 60,000; Year 4 = 65,000; Year 5 = 70,000). One time development costs were RM 90,000 and recurring costs beginning in Year 1 were RM 40,000 over the duration of the system's life. The discount rate for the company was 10 percent. Using a five-year horizon, calculate the net present value of these costs and benefits. Also calculate overall return on investment of the project and then present a break even analysis. At what point does breakeven occur?

Problems							
	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Totals
Net Economic Benefit							
Discount Rate (%)							
PV of Benefits							
NPV of all Benefits							
One-time Costs							
Recurring Costs							
Discount Rate (%)							
PV of Recurring Costs							
NPV of all Costs							
Overall NPV							
Overall ROI							
Break-even Analysis							
Yearly NPV Cash Flow							
Overall NPV Cash Flow							

4. Assuming monetary benefit of WXY Software Firm starts at the second year onwards, which is RM5350 in the second year, RM 6500 in the third year, and RM7800 in the fourth year. The one-time costs are RM2700. While the recurring costs are RM2200 in the first year and the recurring costs in the second year are 90% of the recurring costs from the first year. Moreover, in the third and fourth year, the recurring costs are RM1800, respectively. The discount rate is 10%. Workout the economic feasibility study by using the concept of Time Value of Money (TVM). Using a four-year horizon, calculate the net present value of these costs and benefits. Also calculate overall return on investment of the project and then present a break-even analysis. At what point does breakeven occur? **Show your answers in two decimal places.**

Problems						
	Year 0	Year 1	Year 2	Year 3	Year 4	Total
Net Economic Benefit						
Discount Rate (%) (4 decimal places)						
PV of Benefits						
NPV of all Benefits						
One-time Costs						
Recurring Costs						
Discount Rate (%)						
PV of Recurring Costs						
NPV of all Costs						
Overall NPV						
Overall ROI						
Break-even Analysis						
Yearly NPV Cash Flow						
Overall NPV Cash Flow						
Project break even occurs between () and () years						
Use first year of positive cash flow to calculate break-even fraction						
Actual break even occurs at						